

Course End Project – Ransomware and Malware: Defense, Analysis, and Response

*Digital Forensics and Steganography: Recovering Hidden Data from a
JPG Image*

By

Sharmila Vijayan

1. Introduction:

This project demonstrates the process of embedding text into a JPG image file using Steganography techniques. Simulating a scenario to delete the file permanently and recover the deleted file from the disk image using FTK Imager and extract the hidden text from the recovered JPG image through digital forensic analysis.

2. Problem Statement:

A financial institution suspects a data breach after suspicious **JPG images** in routine emails suddenly disappear from the system. Although the images appear harmless, unusual network activity alerts the cybersecurity team.

During the investigation, they recover fragments of a deleted image from storage and reconstruct it. The image looks normal, but deeper analysis reveals it contains hidden data that could compromise the organization's security. This discovery prompts the team to continue investigating the attackers and take action to prevent further damage.

3. Project Objective:

The objective of this project is to demonstrate how text can be hidden inside a JPG image using steganography and stored on a two-disk system to simulate real-world data storage.

The file is then deleted to make it inaccessible. **FTK (Forensic Toolkit) Imager** is used to recover the deleted file from the disk image. After recovery, the hidden text is extracted from the JPG image, demonstrating how data can be concealed and later uncovered through digital forensic investigation.

4. Scope of the Project:

The scope of this project focuses on demonstrating digital forensics and steganography techniques used to hide and recover sensitive information from image files. The project will simulate a real-world cybersecurity scenario where hidden data is embedded within a **JPG image** and later recovered through forensic investigation.

This project includes the process of:

- Embedding secret text into a JPG image using a steganography tool such as **SilentEye**.
- The modified image will then be stored in a 2GB NTFS partition within a two-disk system to simulate a realistic storage environment.
- To replicate a potential security incident, the image file will be deleted from the system, making it appear inaccessible.

- The project will then focus on digital forensic recovery techniques using **FTK (Forensic Toolkit) Imager** to analyze the disk image and recover the deleted JPG file.
- Once the file is successfully recovered, the hidden data embedded in the image will be extracted using steganography tools

5. Implementation Details:

- Make sure the disk management is 2GB NTFS partition within a two-disk system to simulate a realistic storage environment.
- Embed a hidden text to the JPG image using **SilentEye** encoding.
- Delete the file from the disk and from the Recycle bin.
- Using FTK Imager, analyze the disk image and recover the deleted image.
- Using **SilentEye** decoding, the hidden text in the JPG image can be extracted.

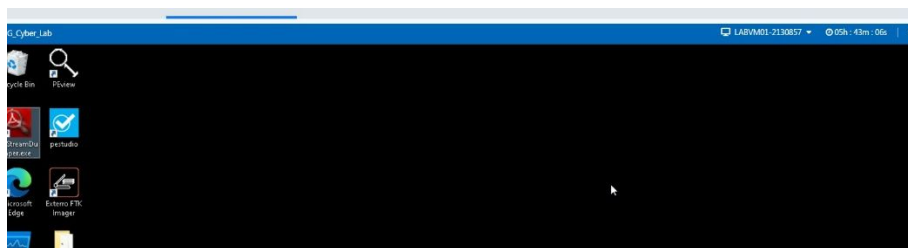
Tools Used:

- SilentEye
- FTK Imager

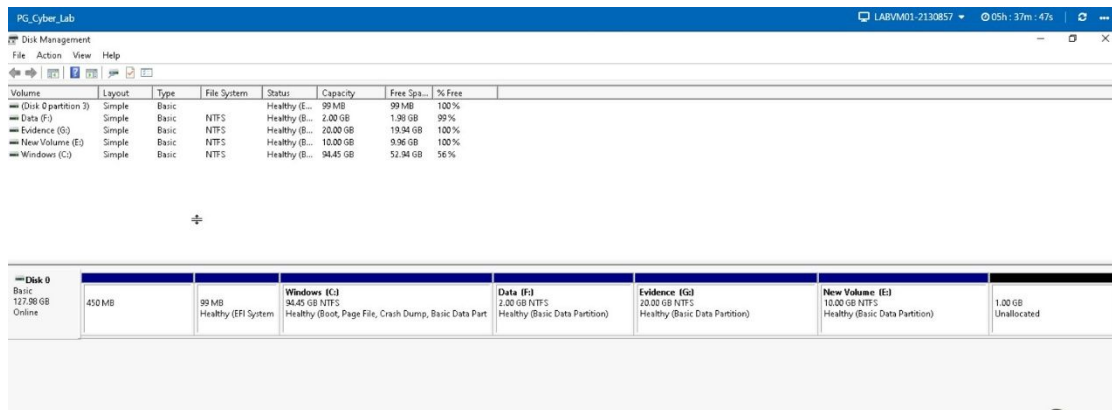
6. Steps to be followed to achieve the results:

Step 1: Disk management should be 2GB NTFS partition within a two-disk system.

1.1 Navigate to KALIVM-1882823

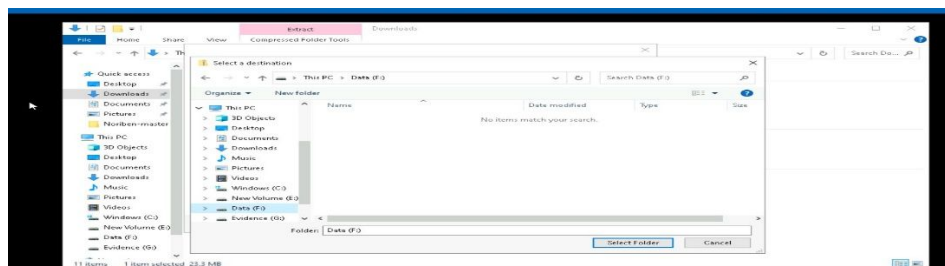


1.2 Right-click the Start button and select Run to open the Run dialog box. Enter diskmgmt.msc and click OK. Confirm that the disk management is 2GB NTFS partition within a two-disk system to simulate a realistic storage environment (Data: (F) and Evidence: (G))

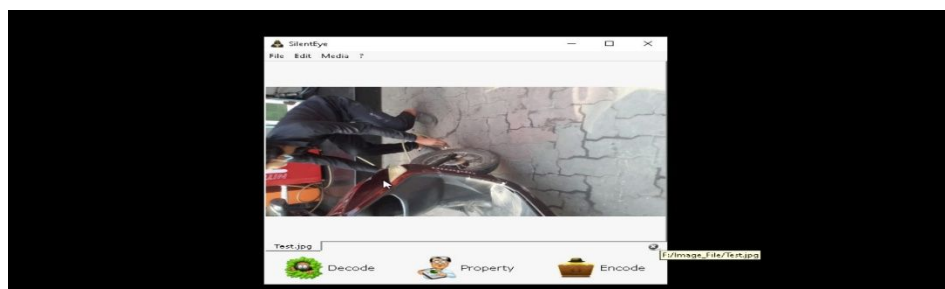


Step 2: Embed a hidden text to the JPG image using SilentEye encoding

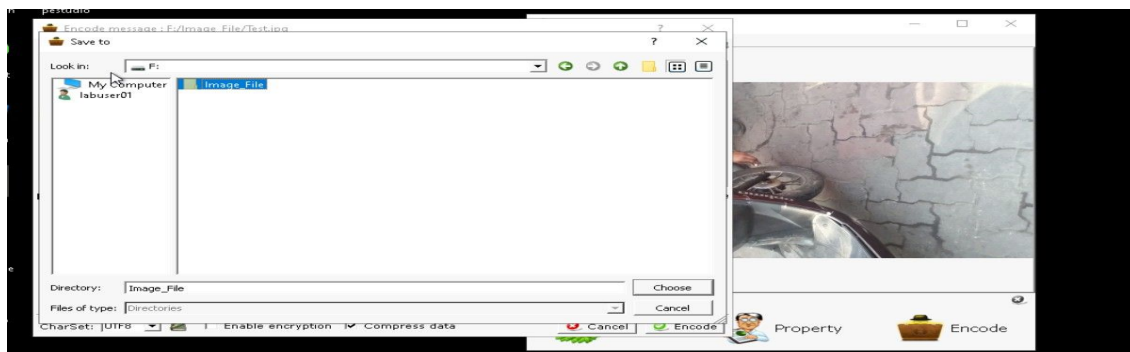
2.1 Download a JPG image “Test.jpg” and save it on Data(:F) disk.



2.2 Open SilentEye tool and open the file “Test.jpg”.



2.3 Add the destination Folder > Create a new Folder “Image_File_new” in Evidence(:G) drive.

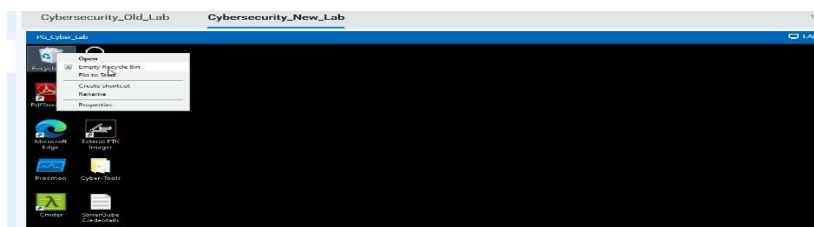


2.4 Enter a message on the “Write a message or select a file to hide” field and click on Encode.



Step 3: Delete the file from the disk and from the Recycle bin.

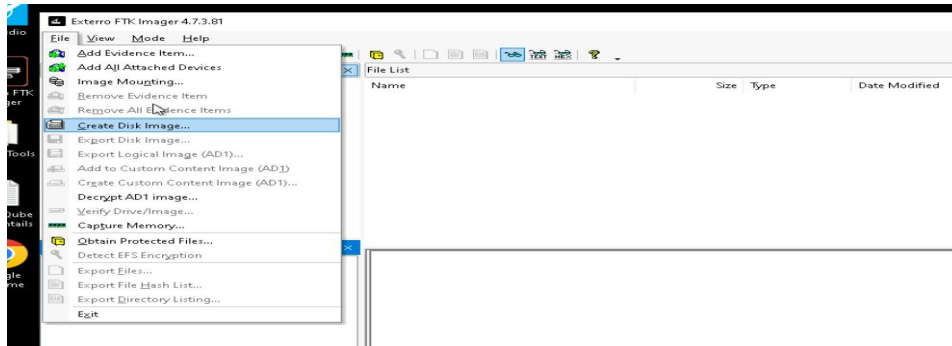
3.1 Go to Drive and delete the JPG image from Drive and Recycle bin.



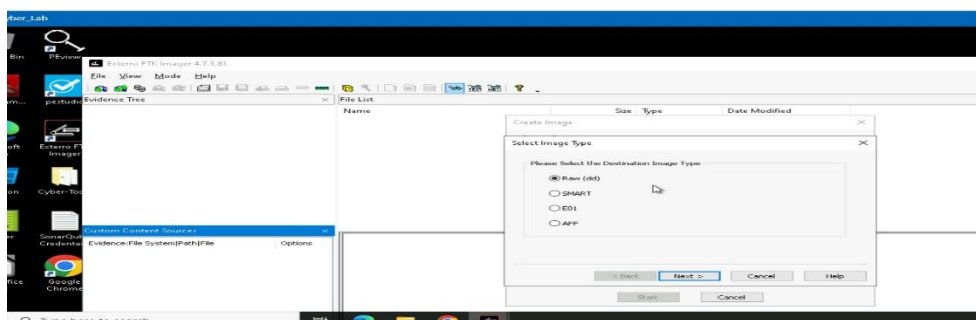
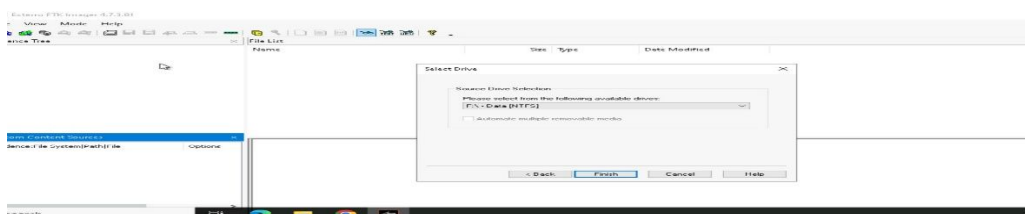
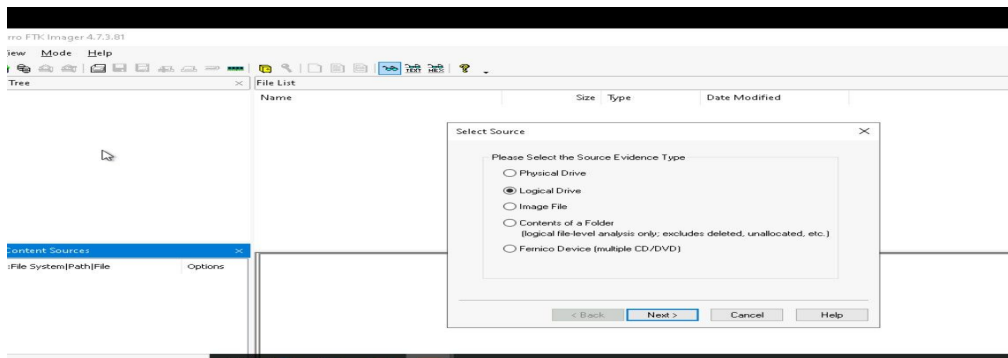
Note: Make sure to shutdown the system and login back.

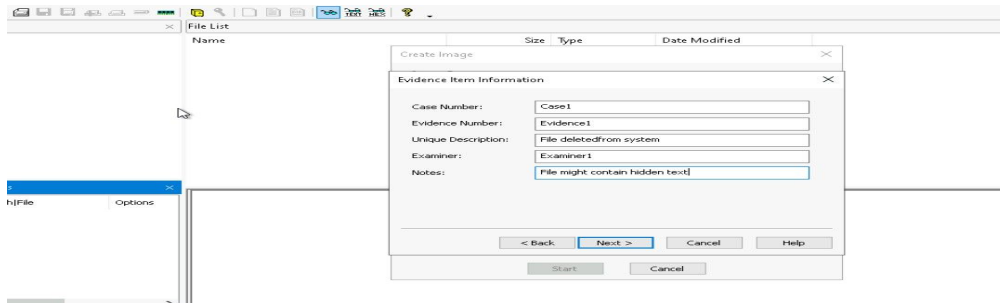
Step 4: Using FTK Imager, analyze the disk image and recover the deleted image.

4.1 Open FTK Imager and File > Create Disk Image.

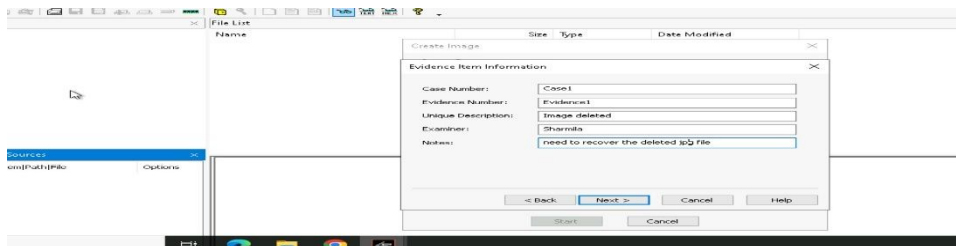


4.2 Select the Source Evidence Type as Logical. Click “Add” button to add the Image Destination and Select the Image type as “Raw”.

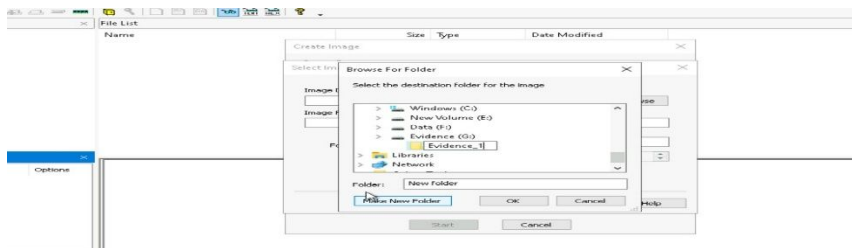




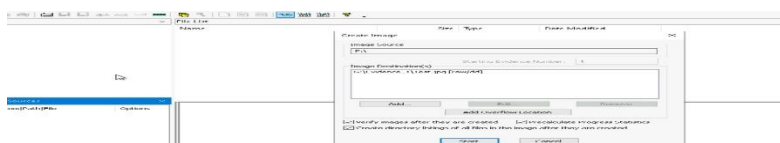
4.3 Enter the Evidence Item information and click Next.



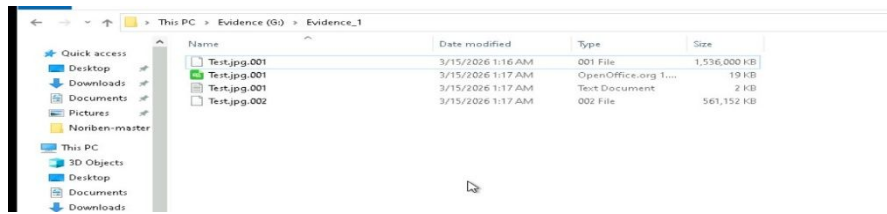
4.4 Click on Image Destination Folder, Create a new folder in Evidence(:G) and add to it. Enter a Image Filename and click Finish.



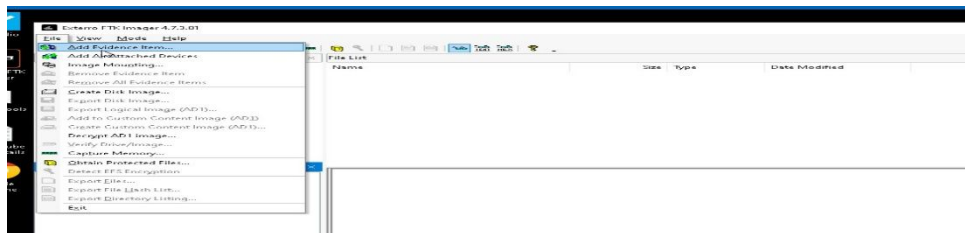
4.5 Click on Start.



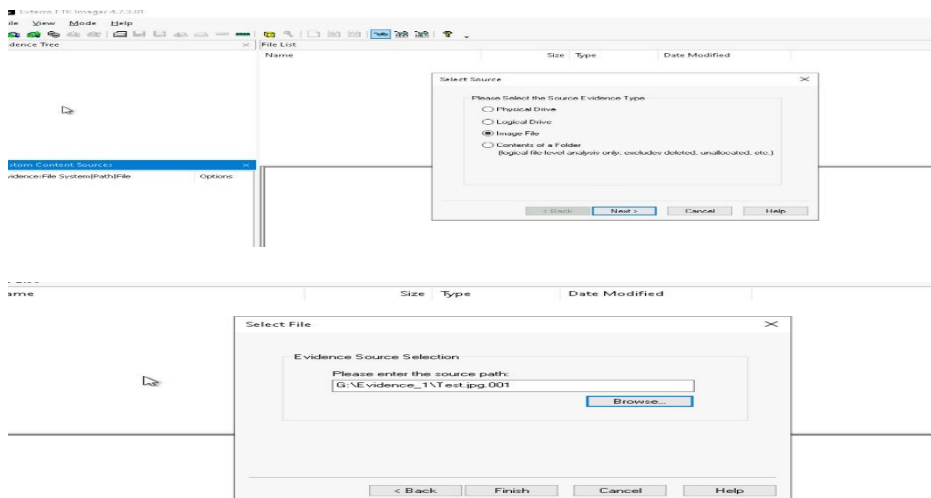
4.6 Go to the Folder Created on Evidence (:G) and verify if files has been copied.



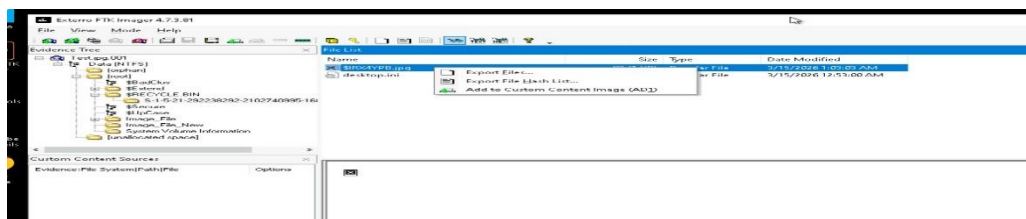
4.7 To access the image> Open FTK Imager and click on Add Evidence Items.

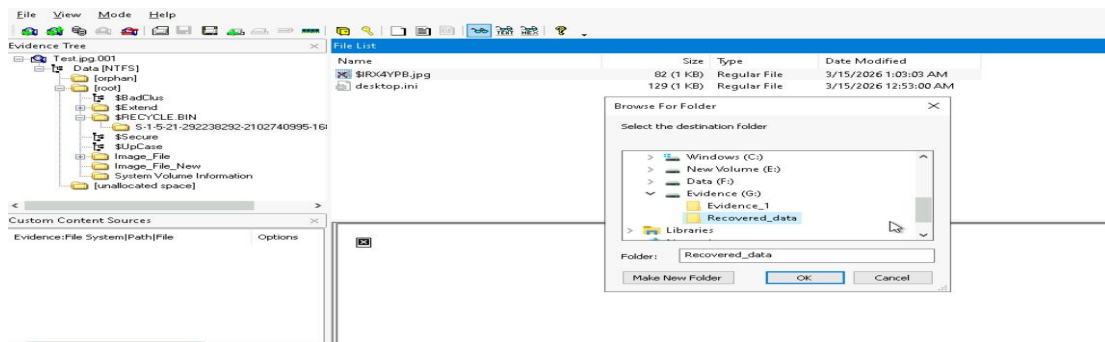


4.8 Select Image > Select the evidence file and click finish.



4.9 Click on file name > navigate to recycle bin > Open the folder> Right Click on the filename > click on extract files > Create a new folder in Evidence (:G) and click OK.

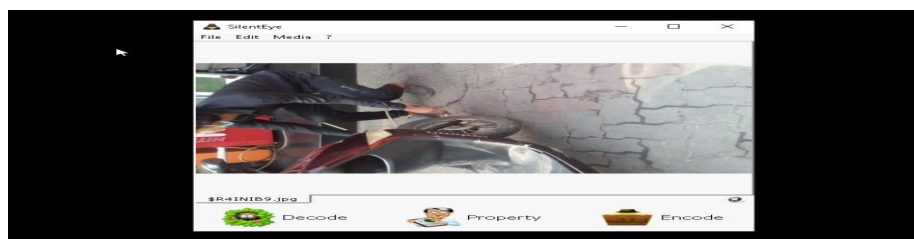
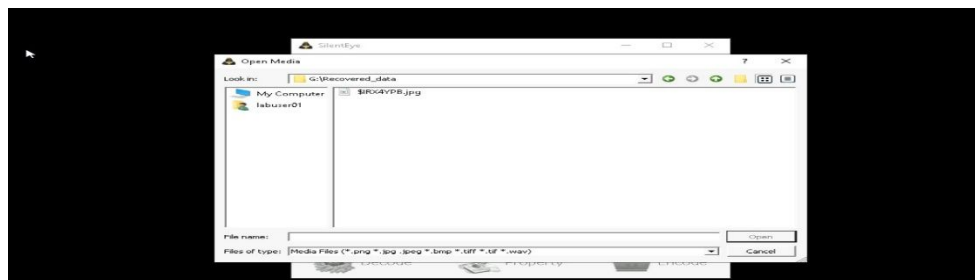




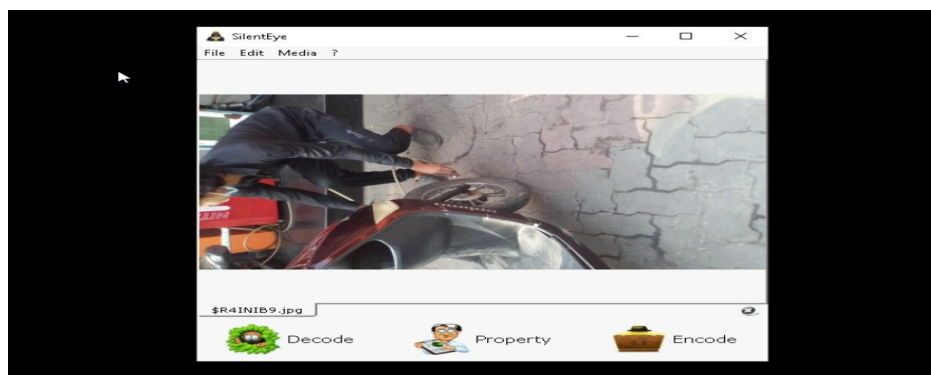
Note: Recovered File is saved in the Evidence folder.

Step 5: Using SilentEye decode the hidden text in the JPG image.

5.1 Open SilentEye Tool and Open the recovered image.



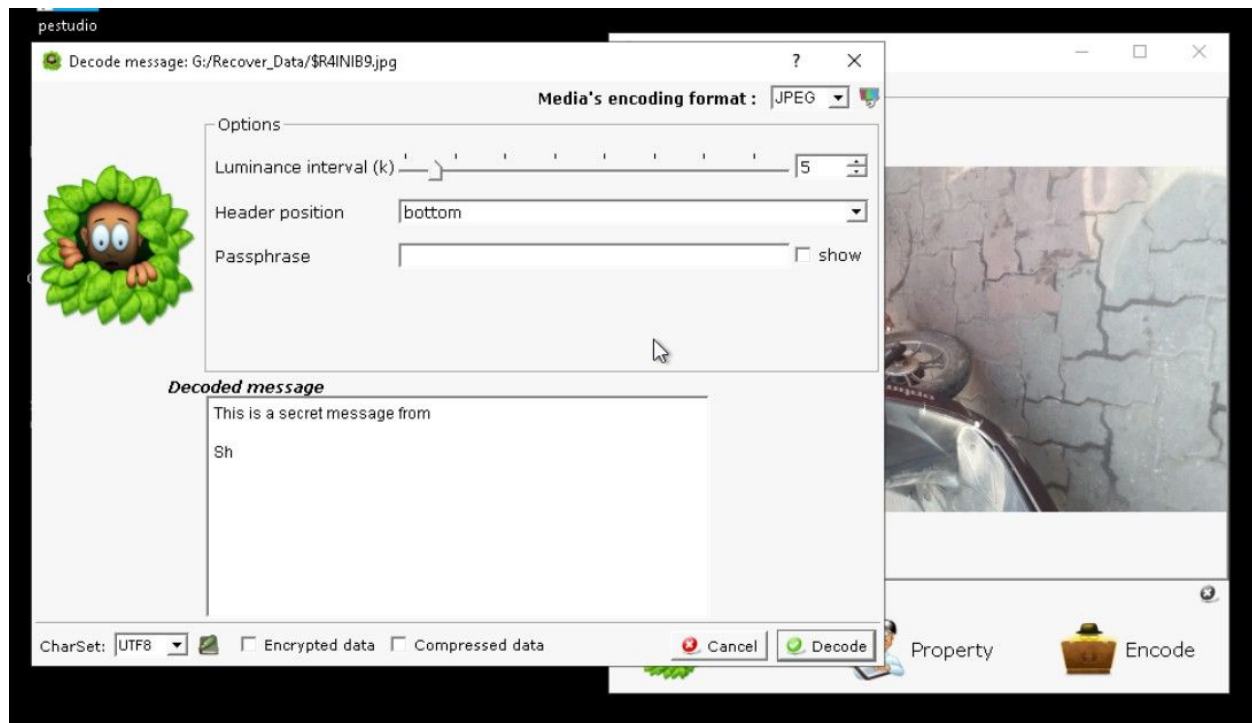
5.2 Click on Decode button. Click on the decode button on the pop up.



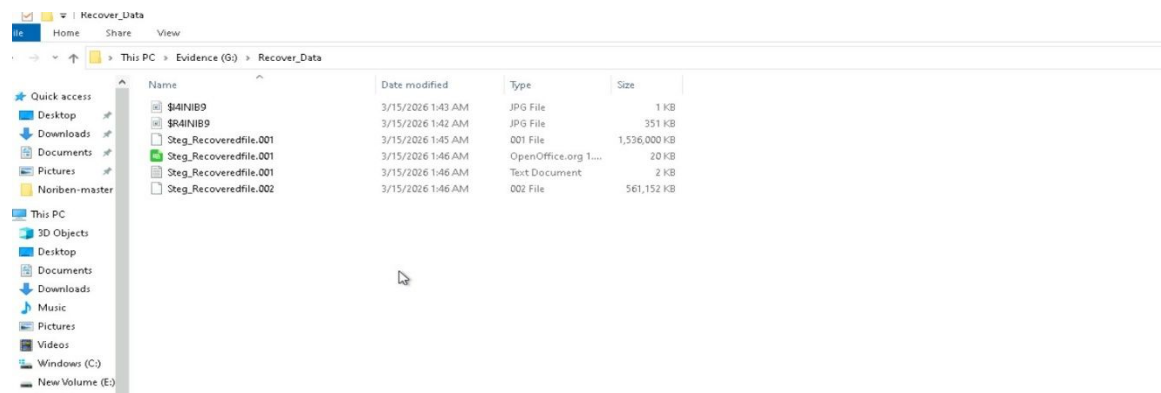


Outputs:

1. Decoded Text in the JPG Image file



2. Recovered Files in the Evidence Drive (: G)



Name	Date modified	Type	Size
\$!4INIB9	3/15/2026 1:43 AM	JPG File	1 KB
\$R4INIB9	3/15/2026 1:42 AM	JPG File	351 KB
Steg_Recoveredfile.001	3/15/2026 1:45 AM	001 File	1,536,000 KB
Steg_Recoveredfile.001	3/15/2026 1:46 AM	OpenOffice.org 1...	20 KB
Steg_Recoveredfile.001	3/15/2026 1:46 AM	Text Document	2 KB
Steg_Recoveredfile.002	3/15/2026 1:46 AM	002 File	561,152 KB

Conclusion:

This project demonstrated how steganography can be used to hide secret data inside a JPG image and how digital forensic tools can recover and analyze deleted files. Using SilentEye, hidden text was embedded in an image, which was later deleted to simulate an attempt to hide evidence. FTK Imager was then used to recover the deleted file and extract the hidden message. The project highlights the importance of digital forensics in detecting hidden data and investigating cybersecurity incidents.

References:

1. Ransomware and Malware: Defense, Analysis, and Response CEP session by Walter

Thank You